

Efforts for Status under Income Inequality

Anurag Srivastava (a.srivastava@pgr.reading.ac.uk)

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Summary

The predominant view of status in empirical economics is of a hierarchy of social ranks. While empirical studies often treat status concerns inherent in observed social ranks separately from economic concerns, the sociological literature also explores the link between status perceptions and economic goals. An important concern for the sociological literature has been to reconcile the subjective aspects of status with its objective aspects. The current chapter focuses on a particular structural basis for a dynamism linking economic and social goals using probability functions that determine the promotion and demotion of participants among rich and poor classes in an economy. The consumer behaviour in response to participation costs and income differences are then explained while considering a promotion probability driven by two characteristics of the consumer - social capital comprising of skills that can be enhanced with expenditure and raw-talents that are unaffected by higher expenditure. Assuming a diminishing returns to utility from wealth optimised by the consumer, we find that a rise in income differences provides more incentive to participate in status competitions. The solution also suggests only two possible states for the economy - one where both rich and poor strive for status and the other where only the rich consumers participate in status competitions.

1 Introduction

While the predominant view of status in empirical economics remains one that is based on hierarchy of ranks (Goldthorpe & Hope (1974); Wegener (1992); Goldthorpe (2000)), several attempts have been made in both sociology and economics to view status in a more dynamic framework (Weber (1925); Shils (1969); Parsons (1971); Coleman (1990); Wegener (1992)). Instead of treating status goals separate from economic goals, the presented chapter pins down a specific link between economics and social goals relevant for status consumption assuming an indirect (instrumental) effect of status on consumer wealth (Cole *et al.* (1992); Postlewaite (1998)).

A contrast between economic and social goals seems has received far more attention in sociological literature (Wegener (1992)) - where the issue of reconciling subjective aspects of status (the effect of status differences on individual behaviour) with its objective aspects (the observed reality of status differences) has been explored in great detail. There seem two broad categories (Wegener (1992)) which the various theories for status can be classified into - those that consider status as a hierarchical order (Parsons (1971); Wegener (1992)) and those that consider it in terms of a conflict among social groups (Weber (1925); Wegener (1992)). Several approaches in economics have also incorporated views from the sociological literature while using a dynamism in the social order relevant for status (Goode (1978); Moldovanu *et al.* (2007)). The current chapter relies upon one such view of status (Goode (1978)) - as a product of economic processes shaped by institutional nurturing or discouragement of individual attributes (also see Moldovanu *et al.* (2007)). Given the extreme variation of economic development that are observed across developing economies (such as Tanzania and Nigeria), the endogenisation of status in such a manner provides us a generic framework to compare disparate developing economies.

Focusing on the specific problem of how variation in urban institutions could influence expenditures such as education that carry some status but also contribute to future income mobility, the proposed chapter evaluates theoretical arguments for how expenditure towards status could respond to changes in income differences and participation costs under urbanisation-related changes. Unlike the typical attribution of status to exogenous factors such as race or caste, the current approach views status under urbanisation in a common Weberian framework contrasting *Gesellschaft* against *Gemeinschaft* in the economy (Weber (1925); Goode (1978); Wegener (1992)).

To specify a structural relationship between the economic and social goals of an individual, the current chapter uses probability functions that define promotion to a richer status class. While the probability of promotion for the consumer is driven by her expenditure, the utility for the consumer comes only from her accumulated wealth. Our approach where participants do not consider the rich status directly into the utility (and therefore do not directly maximise

the chances to be rich) - follows Postlewaite (1998) closely who also discuss the relevance of uncertainty in conspicuous consumption. In the probabilistic game we use to represent status competitions, the participants compete both with raw-talent and acquired-skills (social capital) to get promoted to (or to maintain) a rich position. More specifically, the winner of the game is promoted to a rich position (or maintains her rich position) by having the maximum total score comprising both of raw-talent r_i (uniformly distributed in a population) and social capital s_i (enhanced with expenditure). The social-capital s_i that is enhanced through expenditure is also subject to budget constraints imposed on the consumer¹.

It is worth emphasising that the only two types of participants in the tournament view of status used in the current chapter are the poor and rich classes - so that the investments towards status (such as club-membership fees, or higher-education) increase the chances to be promoted to the rich status. Those who don't participate in the tournament certainly suffer with being poor, but those who do spend only benefit probabilistically. A useful analogy to explain such a competition is that of tennis leagues - where rewards are risky but are unlike lotteries in that more expenditure does not linearly increase one's chance to win in a tennis championship. As with the game of tennis, participants with appropriate raw-talent and skills may exist naturally in a population but their success also depends on engagement with other tennis players. More specifically, there may exist biological characteristics such as strength, dexterity, height or psychological abilities (resilience, conscientiousness etc.) that help one's tennis, but there also exist several characteristics such as technique, rankings, knowledge of conventions, racket-customisations that are only available at a significant cost to the participant. The enhancement of these latter skills (social capital) are what the institutional support are meant to represent. It has been argued more generally that the characteristics of status games are often made to reward curated abilities rather than naturally occurring raw-strengths (Goode (1978)).

With the weight μ of raw-talent in the maximum total score considered for the win, the consumer preferences for the probability of $\mu r_i + (1 - \mu)s_i$ (where r_i is raw-talent and s_i are skills that are available with higher expenditure) are considered under the assumption that the participants maintain the population constant over time. The rich participants are thus replaced by either a rich or poor participant only through the tournament that selects highest combined score regardless of rich or poor background of the participant. Since all participants participate in every such tournament, it is possible for a rich participant to lose her rich-position and let another participant acquire her position. We also assume that the preferences for all poor participants are the same as are the preferences of all the rich participants. Having a rich or a poor status by itself only has an indirect effect (through higher expenditure or budget)

¹We do not view leisure as separate from budget constraints in the model. In other words, the leisure relevant for higher status is interpreted simply as a higher cost for the poorer participant. So long as one can view leisure as the amount of money a participant must forgo in order to participate in status competition, the leisure differences can be also viewed as differences in disposable incomes.

on the chance to be promoted (or maintaining one's rich position) - a probability that otherwise depends on the privately known ability r_i and social-capital s_i . All rich participants and all poor participants having the same behaviour also means that the privately known attributes r_i and s_i are not used by the participant to decide upon the expenditure (effort) towards the status game.

Notice that if the rich and poor participants have disparate incomes y_1 and y_2 (so that $y_1 < y_2$), then the above setup evidently puts the poor participants at a disadvantage since they may not afford to pay the costs to enhance their skills (i.e. realise the true worth of their skills s_i due to high participation costs). However, since both raw-talent r_i and social-capital s_i (the latter enhanced by costly institutional support) are randomly distributed in the population, the competition may never rule out the chance for a poor participant to win. For the same reason, the returns from investments to enhance one's skills may also be diminishing since one cannot beat raw-talent and non enhanced-skills after a certain point.

It is important to highlight that above status competitions are not meant to be population-wide tournaments that involve all participants of a society. Instead, the competitions are small-scale with varied locally defined criteria that may have multiple stages. All such competitions are meant to be independent and may vary only in the threshold participation cost and range of distribution of r_i, s_i of the participants. As scope of the presented chapter is limited to the effect of income differences on consumption, we discuss only a particular competition with known distribution of r_i and s_i among a relatively small number of participants.

In summary, a few properties of a game follow where a wealth-optimising participant with a maximum score wins by enhancing her skills (social capital) and using her raw-talent. First, only one player with the maximum score can win a particular tournament. Second, the rank of the participant (rich / poor status) is of no direct significance part of her utility from wealth. Lastly, the privately known raw-talent r_i and skills s_i are of no significance to expenditure decisions - which vary solely on the basis of rich or poor status. A closed-form solution with a diminishing utility from wealth and a threshold level expenditure required for participants to spend in order to avail social capital suggests that the only two equilibria relevant for status consumption are one where rich as well as poor consumer participates in the status competitions and one where only the rich consumer engage in competitions. Further, the higher income differences seem to encourage both poor and rich participants to engage in status competitions while high participation costs discourage the participation from the poor participant. On similar lines with Hopkins & Kornienko (2010), the chapter contrasts equity in income (rewards) against equity in opportunity in the context of status competitions.

In what follows, Section 2 surveys the issues with definition of status in sociology and economics, Section 3 discusses the model and its solution while Section 4 draws some conclusions about expenditure towards status in the growing economies.

2 Literature Survey

Status is often contrasted as a social goal of an individual against her economic goals. Yet the difference between social and economic goals of the individual remains unclear in both sociology and economics (Wegener (1992)). The idea that status is orthogonal to economic goals - or the “pure status” view (see Moldovanu *et al.* (2007)) - seldom fits with the real-world where status both contributes to income and improves with rises in income. Indeed the other extreme view that the economic goals alone motivate status competitions - so that status is derived purely from economic rewards is also unrealistic. As Moldovanu *et al.* (2007) argue, the reality is often somewhere in between the two extremes.

A structural model for the interplay between economic and social goals is yet to specified in either sociology or economics. The dominant view for status in both sociology and economics has been based on hierarchy and social deference (Coleman (1990); Goldthorpe (2000)) - a view that is inherent in occupational status surveys widely used in empirical studies (e.g. the EGP classes or the Goldthorpe class scheme). A fundamental problem that the sociology literature has elaborated further (Wegener (1992)) is the failure in empirical studies to reconcile the difference between objective aspects of status (the observed reality of status differences) and its subjective aspects (the effect of status differences on individual behaviour). An evident issue with addressing only the objective aspect of status using a utilitarian model while leaving the subjective aspect exogenous is that status may never be explained in economic terms when an economy for status is ruled out. As the literature in sociology warns, the idea of social-closure (aggregate or *stände* first discussed by Weber (1925)) ought not be conflated with that of social ranks (Wegener (1992)). The issue is not a mere theoretical one. Empirical studies suggest that the subjective (private) evaluations of status often measure an overall desirability of occupations rather than common objective notions for status (Goldthorpe & Hope (1974)). Adler & Kraus (1985) find that skills or knowledge of individuals to be a better predictor for individual judgments of status. Lastly, the scales of status used in empirical studies are observed to vary significantly across the sections of a society as well (Acker (1980)).

Several approaches have aimed to reconcile the difference between objective and subjective aspects of status (Wegener (1992)). Parsons (1971), for example, addresses the issue by simply ensuring that the subjective and objective aspects coincide in status-related interpretations. Parsons (1971) argues that the basic categories in terms of which we describe a system of hierarchies as a structure is the same as those in terms of which individual behavior or performance should be defined. While culture places a premium on equality, Parsons (1971) argues, the distribution rules in different social subsystems are not aimed at generating equality - thus letting status exist to assert basic importance of equality of membership status while also making allowances for inequalities resulting from equality

of opportunity (Wegener (1992)). The bottom line of these arguments is that societies must institutionalize some balance between equality and inequality at all times (Parsons (1963, 1971)). On the other hand, Goode (1978) discusses an economy of status where status is created as part of the economic processes while social institutions nurture or discourage certain naturally characteristics in a society. Instead of interpreting status items as those that signal hierarchical status, the dynamic view of status allows Goode (1978) to suggest that investments towards status may increase the likelihood of associating with higher-status members of society.

Other attempts to reconcile the subjective and objective aspects of status (and therefore of social-closure and hierarchy) use rational conflict-based arguments to specify the structural basis for status. Coleman (1990), for example, consider power (purportedly amounting to status) as a commodity to be exchanged in the economy - an approach that has drawn much criticism due to the zero-sum characteristics it imposes on status while treating the total status on a status society as fixed (Wrong (1979)). The current chapter instead leans towards the view from Goode (1978) who had argued that as a product of exchange processes, status is likely to be available in ample supply in a society. To specify the processes that link the economic and social goals of the individual, the current chapter uses probability functions leading to hierarchical promotion that depend on an *a priori* distribution of personal attributes. Instead of using status as a wealth-determined goal, the presented chapter follows an approach suggested by Postlewaite (1998) to consider status only of an indirect role in maximisation of consumer utility from wealth. A similar model is implemented by Cole *et al.* (1992) where a participants have a concern for relative standing because of their standing being instrumental in determining ultimate consumption levels rather than being directly contributing to their utility.

The above probability function approach bears some similarity with the Tullock function used in contest theory approaches as well (Tullock (1967)). Further developments in the contest-theory literature relating to multiple stage contests (Parco *et al.* (2005); Stein & Rapoport (2005)) and budget-constraints (Jorgenson *et al.* (1988); Che & Gale (1998, 1997, 1996); Parco *et al.* (2005); Stein & Rapoport (2005); Brusco & Lopomo (2008)) have also explored contexts similar to the model proposed in the current chapter. However, the approach used in the current chapter remains much simpler than that of a multi-stage Tullock contest under budget-constraints. Unlike the setting for two-stage contests explored in the contest theory literature - where the first-stage is used to limit the number of players participating in the second-stage (Baye *et al.* (1993); Amegashie (1999)) - the game presented in the current chapter only uses second-stage to represent the last time-horizon (i.e. end of lifetime) of the participant rather than as a stage to eliminate players for the second-stage. The distribution of budget across the two-stages of the game is also of little concern in the setting explored in the current chapter since it's only the first-stage of the

game when the expenditure can be less than the total budget. More particularly, the expenditure in the second-stage simply amounts to the disposable income y_1 for the poor participant and y_2 for the rich participant. It's only in the first-stage, strictly speaking, therefore that the consumption decisions are made by either participant. More details of the model follow in Section 3.

3 Model

With a privately known raw-talent r_i and skills (social-capital) s_i - that are both uniformly distributed (the latter i.e. social capital enhanced with expenditure) - we consider foremost the likelihood of any participant $i \in [1, N]$ as the probability for their total score $\mu r_i + (1 - \mu)s_i$ being higher than any other participant. Here, $\mu \in (0, 1)$ is a parameter that combines skills s and raw-talent r so that the participant with maximum score $\mu r + (1 - \mu)s$ wins the competition.

Consider that in a much simpler competition with raw-talents alone, the winning participant is one who has the maximum r_i drawn from the sample N . In other words, the winner in a competition of raw-talents is a player j such that $\{r_i < r_j \forall i \in [1, N]\}$ and the probability that a given participant i of raw-skill r_i is the winner is essentially that of all other players $j \neq i$ having ability $r_{j \neq i}$ less than r_i . This is the probability

$$\int_0^1 P(r_i > r_1) \times P(r_i > r_2) \times \dots P(r_i > r_{i-1}) \times P(r_i > r_{i+1}) \dots P(r_i > r_N) dr_i$$

With a uniform distribution of skills $r \in [0, 1]$, we have $P(r_i \geq r) = r_i$ for all $i \in [1, N]$ and therefore, the probability is evaluates to $\int_0^1 (r_i)^{N-1} dr_i = \frac{1}{N}$. With more players, the chances of a particular participant winning evidently declines.

Introducing the skills s_i which are curated in social institutions, we consider a threshold investment $\bar{v}$ which is necessary to be spent for the social capital to be available to a participant. While this is an arrangement that evidently favours the rich participant, there are also diminishing returns to investing $\bar{v}$ towards the richer position.

An important assumption in the foregoing analysis is that raw-talent is less valued than skills cultivated by institutions. As a result, we use the condition $1 - \mu > \mu$ (which is equivalent to the following assumption) in the rest of the discussion.

Assumption 1. $\mu < \frac{1}{2}$

With the threshold investment $\bar{v}$ as a discrete response to costs of participation, the participants face different probabilities to win depending on whether they spend $\bar{v}$ or not. More specifically, the participant takes advantage of the skills with weight $1 - \mu$ only when

she spends an amount $\bar{\nu}$. In the all-or-nothing payment scheme, the participant might avoid the investment altogether and keep the investment towards savings instead. The behaviour of all participants depends only on ν_1, ν_2 - as they are unaware of r_i, s_i before they participate.

The result we use depends on the following distribution function of $x_i \equiv \mu r_i + (1 - \mu)s_i$ under the Assumption 1 (see Appendix A).

$$F_X(x) = P(x_k < x) = P(r_k \mu + s_k(1 - \mu) < x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1 - \mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1 - \mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

In the game where a participant i with the maximum score x_i is the winner, the number of rich positions is fixed to $N - P$ so that only the top-scorer gets to occupy the only rich position available to be filled in every competition. The expected behaviour of the rich and the poor participants in the long-term can be explained with four different cases.

- Case I - The poor participant does not participate due to insufficient investment $\nu_1 = 0$ while the rich participant does participate by spending $\nu_2 = \bar{\nu}$.
- Case II - The rich participant does not participate with $\nu_2 = 0$ but the poor participant does participate i.e. $\nu_1 = \bar{\nu}$.
- Case III - Both poor and rich participants do not participate so that $\nu_1 = \nu_2 = 0$.
- Case IV - Both poor rich and participant participate with $\nu_1 = \nu_2 = \bar{\nu}$.

With Case I, the poor participant can win when her raw-talent (weighted with μ) is higher than the total score of all other poor participants as well as the total score of the rich participants (who compete with acquired skills). The poor are priced out of accessing any skills s in their score in Case I so that only the rich participants with investment $\nu_2 = \bar{\nu}$ have skills $s > 0$. Thus, a given poor participant must be ahead both of all of the poor participants and the rich participants so that a participant i may win only if μr_i is higher than than raw-talent μr_k in all poor participants and also exceeds the score $\mu r_k + (1 - \mu)s_k$ involving skills among the rich participants

$$P_1(\text{win}|r_i) = \prod_{k=1, k \neq i}^P P(\mu r_k < \mu r_i) \prod_{k=P+1}^N P(\mu r_k + (1 - \mu)s_k < \mu r_i) = r_i^{P-1} F_X(\mu r_i)^{N-P} \quad (1)$$

Therefore,

$$P_1(\text{win}|r_i) = \begin{cases} \frac{\mu^{N-P} r_i^{2N-P-1}}{2\mu(1-\mu)} & 0 < r_i < \mu \\ r_i^{P-1} \left(\frac{\mu r_i - \mu/2}{1-\mu} \right)^{N-P} & \mu < r_i < 1-\mu \\ r_i^{P-1} \left(1 - \frac{(\mu r_i - 1)^2}{2\mu(1-\mu)} \right)^{N-P} & 1-\mu < r_i < 1 \\ 0 & \text{otherwise} \end{cases}$$

With the uniform density for r_i (i.e. $f(r_i) = 1$), total probability to win can be calculated as $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) dr_i$. Notice that since only one participant can win the skills-game, the probability $\int_0^1 \int_0^1 P_2(\text{win}|r_i, s_i) ds_i dr_i$ for a rich participant to win is simply $\frac{1}{N-P} - \frac{P}{N-P} \int_0^1 \int_0^1 P_1(\text{win}|r_i, s_i) ds_i dr_i$.

In Case II, when the rich person does not participate (i.e. all rich participants have $\nu_2 = 0$), the poor participant has the skills-advantage in the game so the the probability to win for the poor participant i is

$$\begin{aligned} P(\text{win}|r_i) &= \prod_{k=1, k \neq i}^P P(\mu r_k + (1-\mu)s_k < \mu r_i + (1-\mu)s_i) \prod_{k=P+1}^N P(\mu r_k < \mu r_i + (1-\mu)s_i) \\ &= F_X(x_i)^{P-1} \prod_{k=P+1}^N P(\mu r_k < \mu r_i + (1-\mu)s_i) \quad (2) \end{aligned}$$

In Case III, when neither the rich nor the poor participate in the game, then the game depends only on raw-talents and all participants have the same probability $\frac{1}{N}$ to win the game. This is because $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) dr_i = \frac{r_i^N}{N} \Big|_0^1 = \frac{1}{N}$ where $P(\text{win}|r_i) = \prod_{k=1, k \neq i}^P P(r_k < r_i) \prod_{k=P+1}^N P(r_k < r_i) = r_i^{N-1}$.

In Case IV, when both rich and poor participants set $\nu_1 = \nu_2 = \bar{\nu}$, then the poor participant can win only if her score x_i is above all other participants - rich or poor.

$$\begin{aligned} P(\text{win}|r_i) &= \prod_{k=1, k \neq i}^P P(\mu r_k + (1-\mu)s_k < \mu r_i + (1-\mu)s_i) \prod_{k=P+1}^N P(\mu r_k + (1-\mu)s_k < \mu r_i + (1-\mu)s_i) \\ &= F_X(x_i)^{N-1} \quad (3) \end{aligned}$$

Using symmetry arguments, this winning probability also turns out to be $1/N$ for all the participants.

It is worth emphasising that the effect of expenditure on the winning probability is weakened with higher number of participants in the game where abilities (both skills s_i

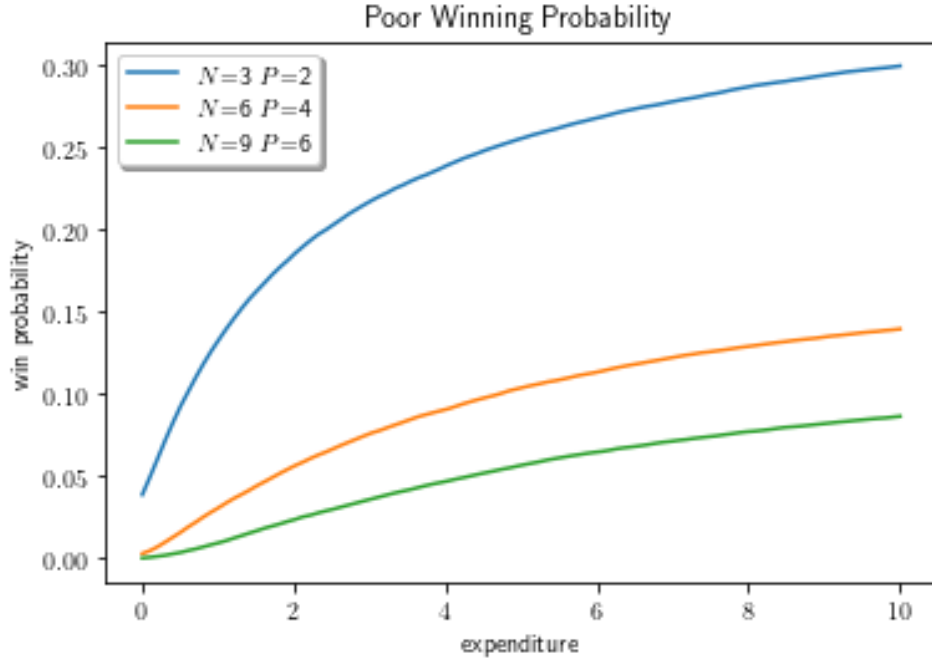


Figure 1: Probability to win for poor with increasing number of participants

relevant for social capital and the raw-talent r_i) are uniformly distributed. In other words, if poor and rich participants were to be considered as two separate groups, then the effect of expenditure on their wins declines when more and more participants are sampled with varied raw-talents and skills. Figures 1 and 2 demonstrate this effect for probability calculated with Monte Carlo simulations for increasing values of the number of poor participants (P) and the total number of participants (N) while hold the ratio P/N constant ($P/N = 2/3$). Our Monte Carlo simulations suggest that the role of expenditure is insignificant for $N \geq 30$.

Since the number of poor is typically higher under real-world conditions, we explore the effect of income-differences using using closed-form solution for $N = 3$ and $P = 2$ in Section 3.1. Equivalent results can be derived using Monte-Carlo simulation for higher values of N and P .

3.1 Equilibrium with $N = 3, P = 2$

We proceed with a solution by setting total number of participants $N = 3$ and the number of poor participants $P = 2$ in Equations 1, 2 and 3 above. As before, a participant takes advantage of the skills with weight $1 - \mu$ only by spending an amount $\bar{v}$ and can decide to avoid the expenditure altogether in order to invest it towards the savings instead. We first discuss the different probabilities in Case I, II, III and IV before discussing the role of an additive intertemporal diminishing utility from accumulated wealth. Recall that the behaviour of both participants depends only on ν_1, ν_2 - as they are unaware of r_i, s_i before they participate.

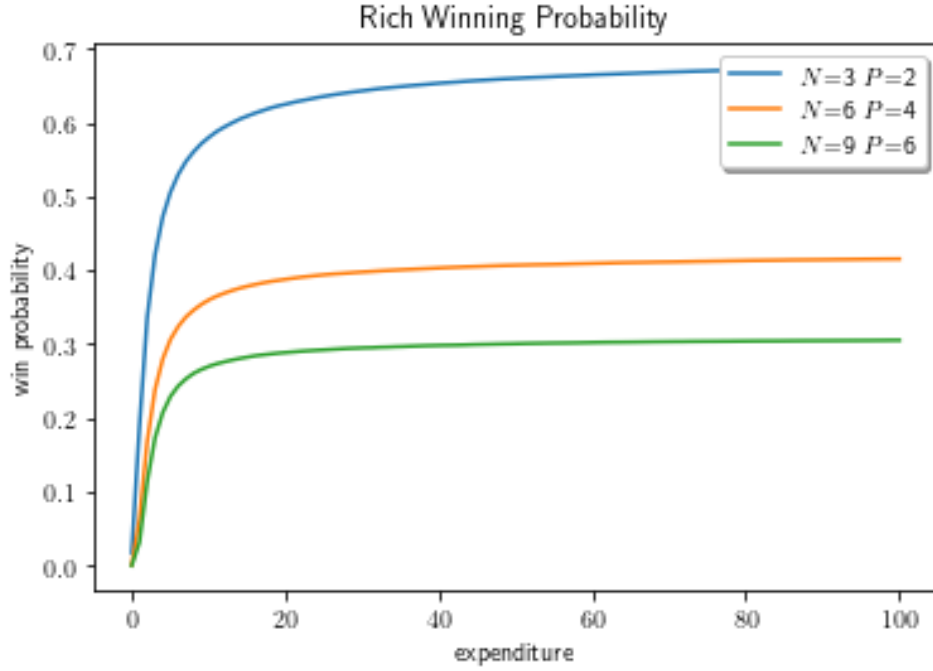


Figure 2: Probability to win for rich with increasing number of participants

An important result for the closed-form solution is the probability distribution (CDF) of $x_i \equiv \mu r_i + (1 - \mu)s_i$ under assumption $\mu < 1 - \mu$ that is detailed in Appendix A.

$$F_X(x) = P(x_k < x) = P(r_k \mu + s_k(1 - \mu) < x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1 - \mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1 - \mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

We now evaluate above probabilities under Case I, II, III and IV discussed above.

Case I

In Case I, the poor participant can win when her raw-talent r_1 is higher than of the total score of rich participant (who competes with acquired skills) as well as the raw-talent (let's say r'_1) of the only other poor participant in the three-player game. Both the poor participants are priced out of accessing any skills s in their score so that the skills $s > 0$ is only true for the rich participant having $\nu_2 = \bar{\nu}$. Thus, the poor participant with raw-talent r_1 can win with the following probability

$$P(\text{win}|r_1) = P(\mu r_2 + (1 - \mu)s_2 < \mu r_1)P(r'_1 < r_1) = F_X(\mu r_1)r_1$$

Notice that $\mu r_1 < \mu < 1 - \mu < 1$ is implied due to $\mu < \frac{1}{2}$ (and $r_1 \in (0, 1)$). The probability that the total score $\mu r_2 + (1 - \mu)s_2$ does not exceed μr_1 implies that only the first part of the piecewise function for $F_X(x)$ needs to be considered. Therefore, using $P(\text{win}|r_1) = \frac{\mu^2 r_1^3}{2\mu(1-\mu)}$, we evaluate $P_1(\text{win})$ as

$$P_1(\text{win}) = \int_0^1 P(\text{win}|r_1)dr_1 = \frac{\mu}{8(1-\mu)}$$

Since only one of the participants can win, the probability $P_2(\text{win})$ for the rich participant to win is simply $1 - 2P_1(\text{win})$. The following thus applies to all other cases (i.e. Cases II, III and IV).

$$P_2(\text{win}) = 1 - 2P_1(\text{win}) \tag{4}$$

Case II

In Case II, the rich person does not participate ($\nu_2 = 0$) and both the poor participants have the skills-advantage in the game. A poor participant wins by having the total score higher than the other poor participant as well as the raw-talent of the rich participant (who does not acquire skills with investment). Representing $x_i \equiv \mu r_i + (1 - \mu)s_i$, we have the following probability for the poor participant to win

$$\begin{aligned} P(\text{win}|r_1, s_1) &= P(\mu r_2 < \mu r_1 + (1 - \mu)s_1) \cdot P(\mu r'_1 + (1 - \mu)s'_1 < \mu r_1 + (1 - \mu)s_1) \\ &= P(\mu r_2 < \mu r_1 + (1 - \mu)s_1)F_X(x_1) \end{aligned}$$

Notice that $P(\mu r_2 < \mu r_1 + (1 - \mu)s_1)$ (i.e. $P(r_2 < r_1 + (\frac{1}{\mu} - 1)s_1)$) is 1 after $r_1 + (\frac{1}{\mu} - 1)s_1 > 1$ (or $\mu r_1 + (1 - \mu)s_1 > \mu$) and $\frac{1}{\mu}$ in the interval $(0, \mu)$ (this is because $r_2 \in (0, 1)$). In other words,

$$P(r_2 < r_1 + (1/\mu - 1)s_1) = \begin{cases} \frac{x_1}{\mu} & 0 < x_1 < \mu \\ 1 & \text{otherwise} \end{cases}$$

The probability to win for the poor participant can be written as (see Appendix A)

$$P(\text{win}|r_1, s_1) = \begin{cases} \frac{x_1^3}{2\mu^2(1-\mu)} & 0 < x_1 < \mu \\ \frac{x_1 - \mu/2}{1-\mu} & \mu < x_1 < 1 - \mu \\ 1 - \frac{(x_1 - 1)^2}{2\mu(1-\mu)} & 1 - \mu < x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

Using double-integral properties (see Appendix B), the total probability to win $P_1(\text{win})$ for a poor participant is evaluated as follows

$$\begin{aligned} P_1(\text{win}) &= \int_{s_1=0}^{s_1=1} \int_{r_1=0}^{r_1=1} P(\text{win}|r_1, s_1) ds_1 dr_1 = \int_{r_1=0}^{r_1=1} \int_{s_1=0}^{s_1=\frac{\mu(1-r_1)}{1-\mu}} \frac{(\mu r_1 + (1-\mu)s_1)^3}{2\mu^2(1-\mu)} ds_1 dr_1 \\ &\quad + \int_{r_1=0}^{r_1=1} \int_{s_1=\frac{\mu(1-r_1)}{1-\mu}}^{s_1=1-\frac{\mu r_1}{1-\mu}} \frac{(\mu r_1 + (1-\mu)s_1 - \mu/2)}{1-\mu} ds_1 dr_1 \\ &\quad + \int_{r_1=0}^{r_1=1} \int_{s_1=1-\frac{\mu r_1}{1-\mu}}^{s_1=1} \left(1 - \frac{(\mu r_1 + (1-\mu)s_1 - 1)^2}{2\mu(1-\mu)}\right) ds_1 dr_1 \\ &= \frac{\mu^2}{10(1-\mu)^2} + \frac{1-2\mu}{2(1-\mu)} + \frac{(4-5\mu)\mu}{8(1-\mu)^2} \\ &= \frac{4\mu^2 + 20(1-2\mu)(1-\mu) + 5(4-5\mu)\mu}{40(1-\mu)^2} = \frac{19\mu^2 - 40\mu + 20}{40(1-\mu)^2} \end{aligned}$$

The probability $P_2(\text{win})$ for the rich participant to win follows from Equation 4.

Case III

In Case III, when neither the rich nor the poor participate in the game, then the game depends only on raw-talents and all participants have the same probability $\frac{1}{3}$ to win the game. This is because $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) dr_i = \frac{r_i^3}{3} \Big|_0^1 = \frac{1}{3}$. For three players, both the poor and rich participants have the same total probability $\frac{1}{3}$ to win.

Case IV

In Case IV, when both rich and poor participants set $\nu_1 = \nu_2 = \bar{\nu}$, then the poor participant can win only if her score x_i is above that of the rich participant as well as that of the only other poor participant. Therefore,

$$P(\text{win}|r_1, s_1) = P(\mu r_2 + (1-\mu)s_2 < \mu r_1 + (1-\mu)s_1)P(\mu r'_1 + (1-\mu)s'_1 < \mu r_1 + (1-\mu)s_1) \\ = F_X(x_1)^2$$

$$P(\text{win}|r_1) = \begin{cases} \frac{x_1^4}{4\mu^2(1-\mu)^2} & 0 < x_1 < \mu \\ \left(\frac{x_1 - \mu/2}{1-\mu}\right)^2 & \mu < x_1 < 1-\mu \\ \left(1 - \frac{(x_1-1)^2}{2\mu(1-\mu)}\right)^2 & 1-\mu < x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

While this integral can be evaluated based on symmetry arguments (since x_i is also randomly distributed), the integral $\int_{s_1=0}^{s_1=1} \int_{r_1=0}^{r_1=1} P(\text{win}|r_1, s_1) dr_1 ds_1$ also evaluates to $\frac{1}{3}$ (see Appendix B)

$$\begin{aligned} \int_{s=0}^{s=1} \int_{r=0}^{r=1} F_X(x_1) ds_1 dr_1 &= \int_{r_1=0}^{r_1=1} \int_{s_1=0}^{s_1=\frac{\mu(1-r)}{1-\mu}} \frac{(\mu r_1 + (1-\mu)s_1)^4}{4\mu^2(1-\mu)^2} ds_1 dr_1 \\ &+ \int_{r_1=0}^{r_1=1} \int_{s_1=\frac{\mu(1-r)}{1-\mu}}^{s_1=1-\frac{\mu r}{1-\mu}} \left(\frac{\mu r_1 + (1-\mu)s_1 - \mu/2}{1-\mu}\right)^2 ds_1 dr_1 \\ &+ \int_{r_1=0}^{r_1=1} \int_{s_1=1-\frac{\mu r}{1-\mu}}^{s_1=1} \left(1 - \frac{(\mu r_1 + (1-\mu)s_1 - 1)^2}{2\mu(1-\mu)}\right)^2 ds_1 dr_1 \\ &= \frac{\mu^3}{24(1-\mu)^3} + \frac{2(1-2\mu)(\mu(7\mu-10)+4)}{24(1-\mu)^3} + \frac{\mu(19\mu^2-30\mu+12)}{24(1-\mu)^3} = \frac{1-\mu^3+3\mu^2-3\mu}{3(1-\mu)^3} = \frac{1}{3} \end{aligned}$$

As is clear from Figure 3 and above results (see summaries in Tables 1 and 2), the condition $\mu < 1/2$ - indicating that curated skills are given more importance than raw-talent in the competitions - means that the poor participant is strictly better off in terms of chances to win (than facing probability $1/3$) if she participates while the rich participant doesn't participate (Case II). Also, the poor participant is better off participating for $\mu < \frac{1}{2}$ when the rich participant does participate as it raises the probability to $1/3$. That the competition becomes uncompetitive both when threshold $\bar{v}$ falls too much (letting both participants participate) or rises too much (letting both participants withdraw) also follows from the properties of the winning probability.

Notice that the rich participant could withdraw and face either $1/3$ or lower probability (when the poor participant participates). By participating, therefore, the rich participant would either face probability $1/3$ (when poor participant also participates) or better (if the poor participant withdraws). Considering the probabilities alone, the rich participant would always participate. If the rich participant always participates, then so long as budget

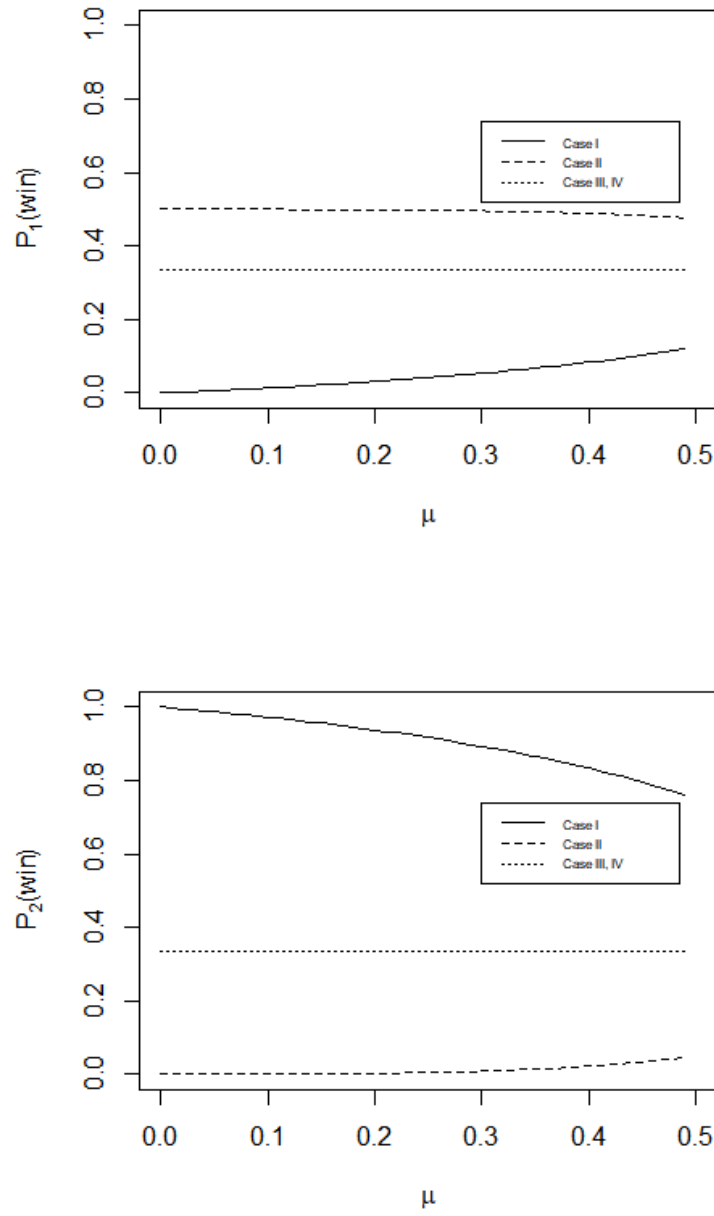


Figure 3: Probability of winning $P_1(\text{win})$, $P_2(\text{win})$ for poor and rich participants with varying μ

constraints permit, the poor participant is also better off participating since not participating brings her probability to win below $1/3$. If the utility was from probability alone, then both participants would participate and face $1/3$ probability.

However, since participants are more likely to face a finite life-time and utility from accumulated assets, they hardly optimise the probability to win in the game. We therefore explore the more likely case where the participants optimise an intertemporally additive utility over their life-time. The life-time in the model is represented with two time-horizons - where the second-time horizon is one before the end of life when all wealth accumulated by either participant must disappear. With no accumulations at the end of life, both rich and poor participants must decide to spend all their incomes y_1 or y_2 (for poor and rich participants respectively) towards the promotion game. In the first-stage, however, the participants decide the amount to be spent towards promotion based upon the utility from wealth they obtain across the two time-horizons that span their lifetime. For the poor participant, the total utility to be optimised (assuming EUT) under the intertemporally additive utility $u(\cdot)$, a winning probability function $W(x)$ (for expenditure x) and a discount-factor δ is

$$u(y_1 - \nu_1) + \frac{1}{\delta}((1 - W(y_1))u(y_1) + W(y_1)u(y_2))$$

Similarly, for the rich participant, the total utility to be optimised (assuming EUT) is

$$u(y_2 - \nu_2) + \frac{1}{\delta}((1 - W(y_2))u(y_1) + W(y_2)u(y_2))$$

With above utilities, we discuss conditions for whether the Case I, II, III, IV can become Nash equilibria (see Table 3). We then proceed to discuss the implications for the effect of income differences on expenditure towards status. It is worth remarking that our discussion of the effect of the threshold $\bar{\nu}$ and the income differences $y_2 - y_1$ on participation (expenditure) assumes that the exogenous parameter μ is stable with respect to short-term changes in threshold investment $\bar{\nu}$ as well as disposable incomes y_1, y_2 .

For ease of notation we define $p_1 \equiv \frac{\mu}{8(1-\mu)}$, $p_2 \equiv 1 - 2p_1$, $p'_1 \equiv \frac{19\mu^2 - 40\mu + 20}{40(1-\mu)^2}$ and $p'_2 \equiv 1 - 2p'_1$ which correspond to the three-player model discussed above. The consumer choices for the poor and rich participants using a general intertemporally additive function $u(A)$ (for assets A) are shown in Table 4. As we can verify from Figure 3 that we have $p_1 < 1/3 < p_2$ and $p'_1 > 1/3 > p'_2$ for all values of $\mu < 1/2$.

Case	Description	Probability of poor participant to win	Response to μ
I	Only rich invest in skills	$\frac{\mu}{8(1 - \mu)}$	With higher μ , the participants with raw-talent get rewarded. The probability to win increases with higher μ for the poorer participant.
II	Only poor invest in skills	$\frac{19\mu^2 - 40\mu + 20}{40(1 - \mu)^2}$	With higher μ , the poor participants with access to skills get rewarded. Therefore, The probability to win decreases with higher μ for the poorer participant.
III,IV	Neither or Both invest in skills	$\frac{1}{3}$	Probabilty to win is independent of μ .

Table 1: The probability of winning for the poor participant in the 3-player model

Given $\mu < 1/2$, we have (5)

$$p_1 < 1/3 < p_2$$

$$p'_1 > 1/3 > p'_2$$

Notice the Table 4 represents the condition when the budget is non-binding i.e. the poor participant is able to spend $\bar{v}$. i.e. $\bar{v} < y_1 < y_2$. When the poor participant faces a binding constraint i.e. $y_1 < \bar{v} < y_2$, then the rich participant knows that the best case probability to win for the poor participant is $\frac{1}{3}$. The rich participant's decision to participate would depend only on utility from savings and the relative advantage from gain in probability. This is elaborated further in Section 3.1.1.

3.1.1 Binding constraint

Assuming EUT participants we first consider the simpler case of binding constraint $y_1 < \bar{v} < y_2$. Since we assume a stability in long-term participant behaviour (which itself is a result of observed probability of success), mixed-strategies are not relevant in the current set up. Considering pure strategies for participation, the rich participant would thus participate

Case	Description	Probability of rich participant to win	Response to μ
I	Only rich invest in skills	$1 - \frac{\mu}{4(1 - \mu)}$	With higher μ , the participants with raw-talent get rewarded. Therefore, the probability to win decreases with higher μ for the rich participant.
II	Only poor invest in skills	$1 - \frac{19\mu^2 - 40\mu + 20}{20(1 - \mu)^2}$	With higher μ , the poor participants with access to skills get rewarded. Therefore, the probability to win increases with higher μ for the rich participant.
III,IV	Neither or Both invest in skills	$\frac{1}{3}$	Probability to win is independent of μ .

Table 2: The probability of winning for the rich participant in the 3-player model

		Poor <i>withdraws</i>	participant <i>participates</i>
Rich	<i>withdraws</i>	Case III	Case II
		Case I	Case IV
Participant		<i>participates</i>	

Table 3: Cases I,II, III, IV as participant strategies

		Poor	participant
Rich participant		<i>withdraws</i>	<i>participates</i>
	<i>withdraws</i>	$u(y_2) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta}$	$u(y_2) + \frac{u(y_2)p'_2 + (1 - p'_2)u(y_1)}{\delta}$
		$u(y_1) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta}$	$u(y_1 - \bar{v}) + \frac{u(y_2)p'_1 + (1 - p'_1)u(y_1)}{\delta}$
	<i>participates</i>	$u(y_2 - \bar{v}) + \frac{p_2u(y_2) + (1 - p_2)u(y_1)}{\delta}$	$u(y_2 - \bar{v}) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta}$
		$u(y_1) + \frac{p_1u(y_2) + (1 - p_1)u(y_1)}{\delta}$	$u(y_1 - \bar{v}) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta}$

Table 4: Utility from two periods for (rich,poor) participants - at the start of the first-period

only when the utility from participating would be higher than from not participating i.e. $u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} > u(y_2) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta}$. This results in the following condition

$$\frac{(p_2 - 1/3)(u(y_2) - u(y_1))}{\delta} > u(y_2) - u(y_2 - \bar{v})$$

With income differences being the same an increase in participation cost $\bar{v}$ discourages participation since RHS increases with $\bar{v}$ ($\because u(y_2 - \bar{v})$ is decreasing in $\bar{v}$) and may fail the above condition. Similarly, with participation cost $\bar{v}$ being the same, an increase in income-difference $y_2 - y_1$ is in the rich participant's interest whereas a narrowing of income differences reduces incentive for the participant to participate by possibly failing the above condition. All else being the same, higher impatience (lower time-discounted value of money or a higher δ) would also discourage participation. Therefore, when budget is binding for the poor participant i.e. if the poor participant cannot participate due to $y_1 < \bar{v}$, then narrower income differences, higher participation costs and higher patience for savings (common for both rich and poor participants) all encourage the rich participant to withdraw from the game. This also means that wider income differences ($y_2 - y_1$) combined with lower participation costs ($\bar{\mu}$) may together help maintain the equilibrium where only the rich participates. Similarly, wider income differences (higher $y_2 - y_1$) and more patience for savings (higher δ) would together maintain the incentive for the rich participant to participate. These conclusions very much fit with the sociological observations made in the literature about status (Hirsch (1977); Galbraith (2001)).

3.1.2 Non-Binding constraint

When budget is non-binding, then participation of the rich participant is her dominant strategy if the condition is also $u(y_2) + \frac{u(y_2)p'_2 + (1-p'_2)u(y_1)}{\delta} < u(y_2 - \bar{v}) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta}$ i.e.

$$\frac{(1/3 - p'_2)(u(y_2) - u(y_1))}{\delta} > u(y_2) - u(y_2 - \bar{v})$$

The inferences on how $y_2 - y_1$, $\bar{v}$ and δ influence the participation (see Section 3.1.1) remain the same for the above condition. To discuss the more general Nash equilibria, we now consider the conditions for all of the Cases I, II, III and IV to become a Nash equilibrium. For ease of notation we define the following for a given $\bar{v}$, y_1 and y_2

$$U_1 \equiv u(y_1) - u(y_1 - \bar{v})$$

$$U_2 \equiv u(y_2) - u(y_2 - \bar{v})$$

One could view U_1 and U_2 as the immediate gain in wealth utility from withdrawing. If $u(\cdot)$ represents a diminishing utility, then it is also necessary for $y_1 < y_2$ that

$$U_2 < U_1$$

Consider now that conditions for Case I to be a Nash Equilibrium. If Case I were to be a Nash equilibrium, then for the rich participant's decision to participate, the poor participant must get more utility from withdrawing than from participating and for poor participant's decision to withdraw, the rich must get more utility from participating than from withdrawing. The following conditions ² must therefore hold simultaneously

$$\begin{aligned} \frac{(\frac{1}{3} - p_1)(u(y_2) - u(y_1))}{\delta} &< U_1 \\ U_2 &< \frac{(p_2 - 1/3)(u(y_2) - u(y_1))}{\delta} \end{aligned} \quad (6)$$

Since $p_2 - \frac{1}{3} > \frac{1}{3} - p_1$, this is a fairly general condition for the equilibrium. An increase in U_1 due to higher $\bar{v}$ does not change the equilibrium conditions so long as the corresponding rise U_2 (due to higher $\bar{v}$) does not affect the second inequality. In other words, a higher $\bar{v}$ favours the equilibrium (Case I) so long as there is an incentive for the rich to participate. Likewise, a decrease in $\bar{v}$ would be favourable for the equilibrium (Case I) so long as the poor participants do not also have an incentive to participate.

Similar effects can be seen with income differences $y_2 - y_1$ (which increase $u(y_2) - u(y_1)$) as well as $\frac{U_2}{U_1}$ if we rewrite Equation 6 as

$$\begin{aligned} \frac{(\frac{1}{3} - p_1)}{\delta} &< \frac{U_1}{u(y_2) - u(y_1)} \\ \frac{U_2}{(u(y_2) - u(y_1))} &< \frac{(p_2 - 1/3)}{\delta} \end{aligned}$$

Limiting our attention to cases when $\bar{v} < y_1$, higher y_2 for fixed y_1 decreases $\frac{U_2}{u(y_2) - u(y_1)}$ and makes it more likely for the rich to participate while it also decreases $\frac{U_1}{u(y_2) - u(y_1)}$ by raising

² $u(y_1) + \frac{p_1 u(y_2) + (1-p_1)u(y_1)}{\delta} > u(y_1 - \bar{v}) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) > \frac{(\frac{1}{3} - p_1)(u(y_2) - u(y_1))}{\delta}$
and $u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} > u(y_2) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) < \frac{(p_2 - 1/3)(u(y_2) - u(y_1))}{\delta}$

the stakes enough for the poor to participate (making it less likely for first inequality to be fulfilled). It is easy to see that higher impatience δ encourages both poor and rich to withdraw. As we shall see, equilibrium conditions corresponding to Case II are far more stringent.

If Case II is a Nash Equilibrium, then for rich's decision to withdraw, the poor participant should get more utility from participating than from withdrawing and for poor participant's decision to participate, the rich participant should get more utility from withdrawing than from participating. This means that the following hold ³ simultaneously,

$$U_1 < \frac{(p'_1 - 1/3)(u(y_2) - u(y_1))}{\delta}$$

$$U_2 > \frac{(1/3 - p'_2)(u(y_2) - u(y_1))}{\delta}$$

Since $\frac{(p'_1 - 1/3)(u(y_2) - u(y_1))}{\delta} < \frac{(1/3 - p'_2)(u(y_2) - u(y_1))}{\delta}$, the above cannot be satisfied so long as we have a diminishing utility which necessitates $U_2 < U_1$. Thus an equilibrium with Case II which incentivises the poor to participate but not the rich is possible only with a reference dependent utility with $U_1 < U_2$. With $U_1 < U_2$, we can write above as

$$\frac{U_1}{u(y_2) - u(y_1)} < \frac{(p'_1 - 1/3)}{\delta} < \frac{(1/3 - p'_2)}{\delta} < \frac{U_2}{u(y_2) - u(y_1)}$$

Notice that despite a reference-dependent utility, a higher $\bar{v}$ or more impatience (δ) may disincentivise the poor from participation just the same (leading to Case III). Similarly, higher income differences (an increase in y_2 for fixed y_1) could encourage the rich to participate by decreasing $\frac{U_2}{u(y_2) - u(y_1)}$ (leading to Case IV). Even with the reference dependent utility, the equilibrium with Case II remains a fragile one.

If Case III were to be a Nash-Equilibrium, then for rich participant's decision to withdraw, poor should get more utility from withdrawing than from participating and for poor participant's decision to withdraw, the rich participant should get more from withdrawing than from participating. The following should thus hold⁴,

$$^3 u(y_1 - \bar{v}) + \frac{p'_1 u(y_2) + (1 - p'_1)u(y_1)}{\delta} > u(y_1) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) < \frac{(p'_1 - 1/3)(u(y_2) - u(y_1))}{\delta}$$

$$\text{and } u(y_2) + \frac{p'_2 u(y_2) + (1 - p'_2)u(y_1)}{\delta} > u(y_2 - \bar{v}) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) > \frac{(1/3 - p'_2)(u(y_2) - u(y_1))}{\delta}$$

$$^4 u(y_1) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta} > u(y_1 - \bar{v}) + \frac{p'_1 u(y_2) + (1 - p'_1)u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) > \frac{(p'_1 - \frac{1}{3})(u(y_2) - u(y_1))}{\delta}$$

$$\text{and } u(y_2) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta} > u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1 - p_2)u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) > \frac{(p_2 - \frac{1}{3})(u(y_2) - u(y_1))}{\delta}$$

$$U_1 > \frac{(p'_1 - \frac{1}{3})(u(y_2) - u(y_1))}{\delta}$$

$$U_2 > \frac{(p_2 - \frac{1}{3})(u(y_2) - u(y_1))}{\delta}$$

Since $p'_1 - \frac{1}{3} < p_2 - \frac{1}{3}$, we have $\frac{(p'_1 - \frac{1}{3})(u(y_2) - u(y_1))}{\delta} < \frac{(p_2 - \frac{1}{3})(u(y_2) - u(y_1))}{\delta}$ and $U_2 < U_1$. A higher $\bar{v}$ pushes both U_2 and U_1 further away from participation but a fall in $\bar{v}$, incentivises the rich participant to participate (leading to Case I).

$$\frac{U_1}{u(y_2) - u(y_1)} > \frac{(p'_1 - \frac{1}{3})}{\delta}$$

$$\frac{U_2}{u(y_2) - u(y_1)} > \frac{(p_2 - \frac{1}{3})}{\delta}$$

Higher income differences i.e. higher y_2 for fixed y_1 also decrease $\frac{U_2}{u(y_2) - u(y_1)}$ and incentivise the rich participant to participate (leading to an equilibrium with Case I). Conversely, if the game is not lucrative for the rich it would automatically be infeasible for the poor i.e. if rich withdraws then poor automatically withdraws. If there is any net utility from playing the game, however, Case IV is the likely equilibrium.

If Case IV were to be a Nash Equilibrium, then for rich's decision to participate, the poor participant should get more from participating than from withdrawing and for poor's decision to participate, the rich participant should get more from participating than from withdrawing. Therefore, the following⁵ must hold

$$U_1 < \frac{(\frac{1}{3} - p_1)(u(y_2) - u(y_1))}{\delta}$$

$$U_2 < \frac{(\frac{1}{3} - p'_2)(u(y_2) - u(y_1))}{\delta}$$

With $\frac{(\frac{1}{3} - p'_2)(u(y_2) - u(y_1))}{\delta} < \frac{(\frac{1}{3} - p_1)(u(y_2) - u(y_1))}{\delta}$ and $U_2 < U_1$, this equilibrium rests upon the incentive for the poor participant to participate. So long as the poorer participant finds the competition worthwhile, the rich would also participate.

The above results outline how participation costs influence the investments towards status. For very low $\bar{v}$, rich and poor would both participate and for very high $\bar{v}$, both would withdraw. For everything in between, the only equilibrium with a diminishing utility from wealth is one

⁵ $u(y_1 - \bar{v}) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta} > u(y_1) + \frac{p_1 u(y_2) + (1-p_1)u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) < \frac{(\frac{1}{3} - p_1)(u(y_2) - u(y_1))}{\delta}$
and $u(y_2 - \bar{v}) + \frac{\frac{1}{3}u(y_2) + \frac{2}{3}u(y_1)}{\delta} > u(y_2) + \frac{p'_2 u(y_2) + (1-p'_2)u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) < \frac{(\frac{1}{3} - p'_2)(u(y_2) - u(y_1))}{\delta}$

where rich participate whereas poor don't.

It's worth highlighting that these results are not limited to a discrete response cost-function and that more general cost-functions also confirm a similar effect of $\bar{v}$. Unlike the discrete response function where a participant misses out on skills completely if she does not spend a certain threshold payment, we run Monte Carlo simulations with a skills-multiplier $c(x)$ for social-capital which lets every participant i have the score $\mu r_i + (1 - \mu)c(x)s_i$ so that skills s_i are realised fully ($c(x) = 1$) only at limit. As with the discrete response, skills are 0 for no expenditure/effort in this continuous response cost-function. Here too, we find that the poor participant backs out from (reduces) expenditure while the rich increases her expenditure in the Nash equilibrium.

The above framework also allows us to discuss the implications of inequality for the same total wealth in the economy. As such, a segregation where the poor never participate in the status economy seems less favourable than one where both rich and poor participate in the status competitions. The current chapter explains that the likely outcome of a failure to maintain universal participation in status games (under diminishing utility from wealth) is that of a segregation where only the rich participate in skills leading to economic success. While a reference-based utility (which may be encouraged with segregation) does seem to permit an equilibrium where only the poor are concerned with status games (with respect to skills-enhancement), the result only suggests that a segregation of status games along with that of incomes is another possible long-term outcome of limited participation.

4 Conclusion

We have considered a model where status amounts to a position that only temporarily cements the participant's economic position. With a decision theory framework for uncertainty, the goal of the model has been to investigate role of income difference on expenditures that carry some status while also having some effect on the income mobility of the participants. The endogenisation of status in this manner allows us to use the framework to understand the role of income difference under urbanisation across economies with wide-differences in status-determining cultural notions as well as development indicators.

The foremost result from the model is that under diminishing utility from wealth, the only two stable equilibria are those where both rich and poor participate in status competitions and those where only rich stand to benefit from status competitions. The model allows us to understand how a withdrawal from expenditure such as education in certain developing economies could be result of a combined effect of both participation costs and income-differences. More specifically, high participation costs push the equilibrium to one where only rich participate in status-related competitions whereas

high income differences may have the unlikely effect of incentivising the poor consumers to participate. In the context of developing economies, lowering of participation costs in education (as a status game) despite rise in income differences may thus also represent a likely escape from segregation of participation in the near-term for many developing economies. The role of a reference-based utility add an important caveat to these results since contentment at lower levels of wealth could support segregation of competitions at different income levels in the economy.

Appendix A

The distribution of a convex combination $w = \mu r + (1 - \mu)s$ of two random variables r, s - which are distributed according to densities $f_R(r)$ and $f_S(s)$ - can be derived using the convolution of two derived variables $x = \mu r$ and $y = (1 - \mu)s$. If r and s are uniformly distributed variables ranging from 0 to 1, then x ranges from 0 to μ and y ranges from 0 to $(1 - \mu)$. The CDF for $F_X(x)$ and $F_Y(y)$ are simply $F_X(x) = \frac{x}{\mu}$, $F_Y(y) = \frac{y}{1-\mu}$ (respectively). Therefore, the densities for x and y are $f_X(x) = \frac{1}{\mu}$ and $f_Y(y) = \frac{1}{1-\mu}$. The density of w is simple $f(w) = \int_{-\infty}^{\infty} f_Y(w - x)f_X(x)dx$. Since, $f_X(x)$ is $1/\mu$ only between 0 and μ . We have the following integral to evaluate for the convolution

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w - x)dx$$

Since $f_Y(y)$ is 1 only between 0 and $1 - \mu$, we have the condition $0 < w - x < 1 - \mu \Rightarrow w - (1 - \mu) < x < w$. Further, x is distributed only between 0 and μ . As shown in Figure 4, the integral would evaluate to the following⁶ under the assumption $\mu < \frac{1}{2}$.

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w - x)dx = \begin{cases} \frac{1}{\mu} \int_0^w \frac{1}{1-\mu} dx & 0 < w < \mu \\ \frac{1}{\mu} \int_0^{\mu} \frac{1}{1-\mu} dx & \mu < w < 1 - \mu \\ \frac{1}{\mu} \int_{w-(1-\mu)}^{\mu} \frac{1}{1-\mu} dx & 1 - \mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

This can be simplified as

$$f(w) = \begin{cases} \frac{w}{\mu(1-\mu)} & 0 < w < \mu \\ \frac{1}{1-\mu} & \mu < w < 1 - \mu \\ \frac{1-w}{\mu(1-\mu)} & 1 - \mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

Figure 5 shows this trapezoidal density with μ set to $\frac{1}{5}$. The CDF $F_X(x)$ can be evaluated by integrating the above density as $P(X < x) = \int_{-\infty}^x f(t)dt$.

⁶Notice that as a convex combination, w is distributed only between 0 and 1.

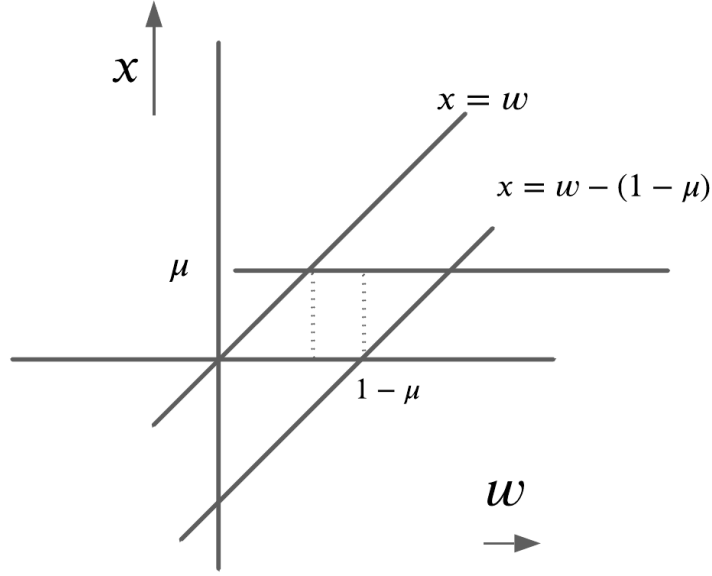


Figure 4: Ranges of w and $w - x$ for convolution of two scaled random variables

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{\mu}{2(1-\mu)} + \frac{x-\mu}{1-\mu} & \mu < x < 1-\mu \\ \frac{\mu^2}{2\mu(1-\mu)} + \frac{2\mu(1-2\mu)}{2\mu(1-\mu)} + \frac{\mu^2 - x^2 + 2x - 1}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Simplifying above, we have

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

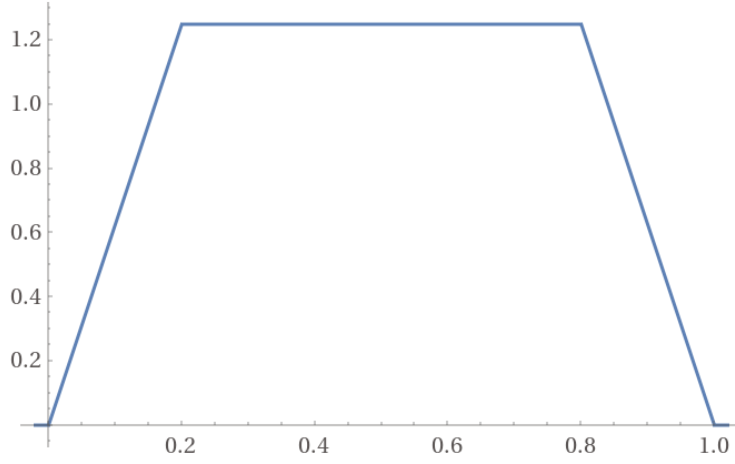


Figure 5: Combined density of w as the convex combination of two random variables with $\mu = \frac{1}{5}$

Appendix B

To calculate $\int_{s=0}^{s=1} \int_{r=0}^{r=1} P(\text{win}|r, s) dr ds$ over the variable x (where $0 < r < 1$ and $0 < s < 1$), we evaluate the sum of three parts split by the boundaries of the piecewise function. The three parts are shown in Figure 6. If $x = \mu r + (1 - \mu)s$ and $F(x)$ is of the following piece-wise form

$$F(x) = \begin{cases} A & 0 < x < \mu \\ B & \mu < x < 1 - \mu \\ C & 1 - \mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Then the integral $\int_{s=0}^{s=1} \int_{r=0}^{r=1} F(x) dr ds$ can be written as

$$\int_{s=0}^{s=1} \int_{r=0}^{r=1} F(x) ds dr = \int_{r=0}^{r=1} \int_{s=0}^{s=\frac{\mu(1-r)}{1-\mu}} A ds dr + \int_{r=0}^{r=1} \int_{s=\frac{\mu(1-r)}{1-\mu}}^{s=1-\frac{\mu r}{1-\mu}} B ds dr + \int_{r=0}^{r=1} \int_{s=1-\frac{\mu r}{1-\mu}}^{s=1} C ds dr$$

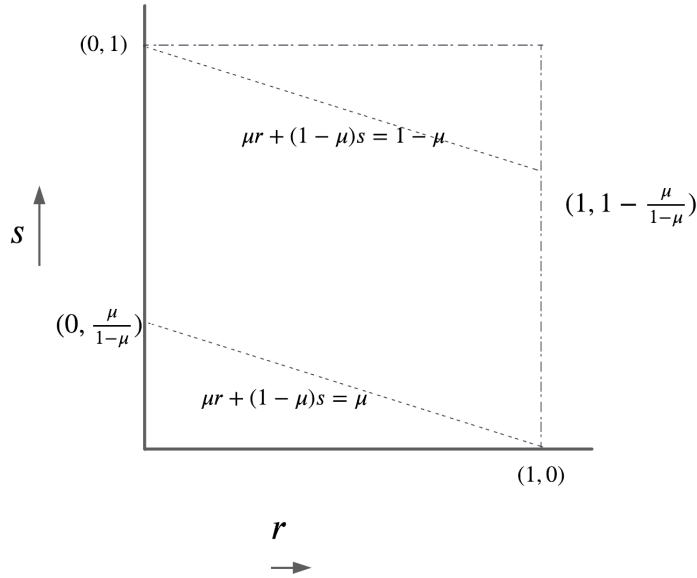


Figure 6: Integration over r, s split by boundaries $\mu r + (1 - \mu)s = \mu$ and $\mu r + (1 - \mu)s = 1 - \mu$

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